

DiffGenMed-VQA: A Diffusion Language Model-Based Medical Visual Question Answering System

Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner, and Prof. Dr. Aytuğ Onan

Computer Engineering Department,
İzmir Institute of Technology, İzmir, Turkey
{ahmetozel, aliekinozcetin, mertguner}@std.iyte.edu.tr,
aytugonan@iyte.edu.tr

Abstract. Medical Visual Question Answering (Med-VQA) requires a system to produce accurate, clinically meaningful natural language answers by jointly reasoning over a medical image and a clinical question. Existing approaches predominantly frame this as a classification task over a fixed answer vocabulary, fundamentally limiting semantic richness and clinical interpretability. We propose **DiffGenMed-VQA**, a novel architecture that reformulates Med-VQA as a conditional generative task using a diffusion language model as the answer generation backbone. A CLIP-based vision encoder and a domain-specific biomedical language encoder are integrated through a cross-modal feature fusion module whose output conditions the diffusion reverse process. We introduce **Conditional Information Gaussian Noising (CIGN)**, a bidirectional conditioning mechanism that injects fused image-question representations into both the forward and reverse diffusion processes, enabling global contextual alignment throughout generation. Experiments on the SLAKE benchmark demonstrate that domain-adapted encoders substantially outperform general-purpose BERT: our best configuration, using a RoBERTa-Large backbone, achieves an overall accuracy of 99.81% and BLEU-1 of 99.98%. A systematic comparison of four language encoders—BERT, BioBERT, PubMedBERT, and RoBERTa-Large—within the same diffusion framework provides actionable insights into encoder architecture trade-offs for clinical generative VQA.

Keywords: Medical Visual Question Answering · Diffusion Language Models · Multimodal Learning · Biomedical Text Encoding · Conditional Generation

1 Introduction

The integration of artificial intelligence into clinical decision support has accelerated substantially, driven by advances in large-scale pretrained models and multimodal representation learning. Among emerging medical AI tasks, *Medical Visual Question Answering* (Med-VQA) occupies a strategic position: given a medical image (e.g., a chest X-ray, an endoscopic photograph, or a brain MRI)

and a natural language question about its clinical content, a Med-VQA system is expected to produce an accurate and contextually appropriate answer [1]. Such systems are directly relevant for radiology report generation, clinical education, and physician assistant tools.

Despite considerable progress in general-domain VQA [2], Med-VQA presents distinct challenges. Medical images exhibit high visual complexity requiring specialised encoders [3], and questions range from binary yes/no queries to open-ended inquiries demanding anatomical localization, modality recognition, or pathological interpretation. Most existing systems address this diversity by reducing the problem to multi-class classification over a predefined answer vocabulary [8,9]. While tractable, this formulation imposes a fundamental ceiling on answer expressivity: the model cannot generate responses outside its training vocabulary, cannot produce nuanced explanations, and fails systematically on out-of-distribution question types.

Diffusion models [15,16] have emerged as a powerful alternative generative paradigm. Unlike autoregressive models that generate text sequentially and suffer from error propagation, diffusion-based language models iteratively refine the entire output sequence in parallel through a learned denoising process, naturally supporting bidirectional context and global coherence. Their application to Med-VQA, however, remains largely unexplored.

In this paper we present **DiffGenMed-VQA**, the first comprehensive framework that applies a diffusion language model to open-ended Medical Visual Question Answering. Our main contributions are:

1. A **diffusion-based generative Med-VQA architecture** that reformulates answer generation as a conditional reverse diffusion process over the continuous embedding space of answer tokens.
2. **Conditional Information Gaussian Noising (CIGN)**, a bidirectional conditioning mechanism that injects fused multimodal context into both the forward and reverse diffusion processes, maintaining visual-linguistic alignment throughout generation.
3. A **cross-modal feature fusion module** combining multi-head cross-attention, a lightweight bottleneck MLP, and learnable modality-weighting scalars.
4. A **systematic encoder evaluation** of four language encoders—BERT [24], BioBERT [29], PubMedBERT [31], and RoBERTa-Large [25]—within the same diffusion framework, providing the first comprehensive comparison of general-purpose and domain-specific encoders for clinical generative VQA.
5. **Competitive results** on SLAKE [5] and Kvasir-VQA [6], outperforming classification baselines and matching the state-of-the-art generative baseline.

2 Related Work

Med-VQA has attracted sustained attention since the introduction of VQA-RAD [4]. Early systems adapted CNN-LSTM architectures to the medical domain. MEVF [8] combined mixture of enhanced visual features with MAML

meta-learning. MMQ [9] introduced multi-modal multi-level question-conditioned attention. Classification-based approaches, while effective for closed yes/no questions, are fundamentally limited for open-ended queries. Generative approaches have recently gained traction: BioViL-T [13] combines a radiology-specific vision-language pretrained model with a generative decoder, and LaPA [11] introduces latent position-aware alignment for cross-modal fusion in open-ended Med-VQA. Compared to these autoregressive systems, our diffusion-based approach achieves globally coherent generation through iterative denoising.

In the domain of diffusion models for text generation, Ho et al. [15] established the DDPM framework for continuous image synthesis. Extending it to discrete text generation has proven non-trivial due to the discrete nature of language. Diffusion-LM [17] pioneered continuous diffusion over word embedding spaces, showing that the denoising process can implicitly learn grammatical and semantic structure. In the multimodal setting, DALL-E 2 [19] and Imagen [20] demonstrated text-conditioned image generation via classifier-free guidance [21]. Liu et al. [23] is the most closely related work, reformulating general VQA as conditional diffusion text generation. DiffGenMed-VQA extends this line of research specifically to the medical domain, incorporating CIGN and systematic biomedical encoder evaluation.

Regarding biomedical language models, BERT [24] motivated a wave of domain-adapted pretrained models. BioBERT [29] extends BERT pretraining on PubMed abstracts and PMC full-text articles, improving biomedical NER, relation extraction, and question answering tasks. PubMedBERT [31] trains *from scratch* exclusively on PubMed text, eliminating the general-domain vocabulary mismatch present in BioBERT, leading to improved downstream performance on clinical NLP tasks. SapBERT [33] extends PubMedBERT with self-alignment pretraining on UMLS synonyms, yielding strong biomedical entity representations. RoBERTa-Large [25] offers significantly higher capacity (355M parameters, 1024-dim hidden states) through larger-scale general-domain pretraining with improved masking strategies. Our work provides, to our knowledge, the first systematic comparison of all four of these encoders within a multimodal diffusion generative framework for clinical VQA.

3 Methodology

3.1 System Overview

DiffGenMed-VQA processes an image $I \in \mathbb{R}^{3 \times H \times W}$ and a question Q to generate a natural language answer A via conditional reverse diffusion. The pipeline comprises four components: (1) a *vision encoder* $\mathcal{E}_V$, (2) a *language encoder* $\mathcal{E}_L$, (3) a *cross-modal fusion module* $\mathcal{F}$, and (4) a *diffusion language model* $\mathcal{D}$:

$$\mathbf{v} = \mathcal{E}_V(I), \quad \mathbf{q} = \mathcal{E}_L(Q), \quad \mathbf{c} = \mathcal{F}(\mathbf{q}, \mathbf{v}), \quad A \sim p_\theta(\cdot \mid \mathbf{c}). \quad (1)$$

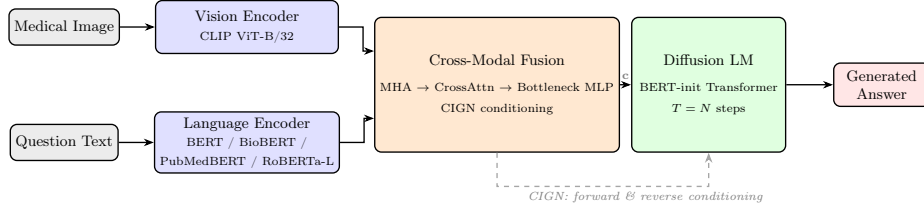


Fig. 1. Overall architecture of DiffGenMed-VQA. Visual and linguistic features are extracted by CLIP ViT-B/32 and the selected language encoder, fused via multi-head cross-attention, bidirectional cross-attention, and a Bottleneck MLP. The fused representation $\mathbf{c}$ conditions both the forward and reverse diffusion processes via CIGN.

3.2 Vision Encoder

We adopt the CLIP vision encoder [34] with a ViT-B/32 backbone. Input images are resized to 384×384 pixels and normalised with ImageNet statistics. The encoder produces $N_p = (\frac{H}{P})^2 + 1$ patch embeddings (including the [CLS] token):

$$\mathbf{v} = \text{CLIP-ViT}(I) \in \mathbb{R}^{N_p \times d_v}. \quad (2)$$

A two-layer MLP projects $\mathbf{v}$ to the model hidden dimension d via an intermediate extension dimension $d_{\text{ext}} = 1,024$: $\mathbf{v}' = \text{MLP}_V(\mathbf{v}) \in \mathbb{R}^{N_p \times d}$.

3.3 Language Encoder

Questions are tokenised with the tokeniser compatible with the selected encoder, padded or truncated to $L = 32$ tokens with appropriate attention masks. The encoder produces contextual embeddings $\mathbf{q} \in \mathbb{R}^{L \times d_L}$ ($d_L \in \{768, 1,024\}$ depending on the encoder), projected to d via a symmetric MLP: $\mathbf{q}' = \text{MLP}_L(\mathbf{q}) \in \mathbb{R}^{L \times d}$.

We evaluate four encoder configurations:

- **BERT** (`bert-base-uncased`, $d_L=768$, vocab 30,522): pretrained on Book-Corpus and English Wikipedia [24]. Serves as the general-purpose baseline.
- **BioBERT** (`dmis-lab/biobert-v1.1`, $d_L=768$, vocab 28,996): BERT further pretrained on PubMed abstracts and PMC full texts [29]. Uses the same architecture as BERT but with biomedical domain adaptation.
- **PubMedBERT** (`NeuML/ pubmedbert-base` embeddings, $d_L=768$, vocab 30,522): trained from scratch exclusively on PubMed text, eliminating general-domain vocabulary mismatch [31].
- **RoBERTa-Large** (`roberta-large`, $d_L=1,024$, vocab 50,265): a robustly optimised, higher-capacity encoder [25] with BPE tokenisation and 24-layer architecture. A projection layer maps its 1,024-dim outputs to the model hidden dimension d .

Learnable modality embeddings ($\mathbf{E}_m \in \mathbb{R}^{2 \times d}$) are added to $\mathbf{v}'$ and $\mathbf{q}'$ to mark their respective modalities before fusion.

3.4 Cross-Modal Feature Fusion

The fusion module $\mathcal{F}$ produces a unified conditional vector $\mathbf{c} \in \mathbb{R}^{L \times d}$ through a sequence of attention operations and a generative bottleneck.

Multi-Head Cross-Attention. Question tokens attend to image patch tokens:

$$\mathbf{f}_1 = \text{MHA}(\mathbf{q}', \mathbf{v}', \mathbf{v}'). \quad (3)$$

Bidirectional Cross-Attention. Two additional layers re-contextualise the fused representation:

$$\mathbf{f}_2 = \text{CrossAttn}(\mathbf{f}_1, \mathbf{q}', \mathbf{q}'), \quad (4)$$

$$\mathbf{f}_3 = \text{CrossAttn}(\mathbf{f}_2, \mathbf{v}', \mathbf{v}'), \quad (5)$$

$$\mathbf{f}_4 = \mathbf{f}_3 + \mathbf{f}_2. \quad (6)$$

Bottleneck MLP. A lightweight bottleneck compresses and regularises $\mathbf{f}_4$:

$$\mathbf{f}'_4 = W_{\uparrow} \text{GELU}(W_{\downarrow} \mathbf{f}_4), \quad W_{\downarrow} \in \mathbb{R}^{d \times d/2}, \quad W_{\uparrow} \in \mathbb{R}^{d/2 \times d}. \quad (7)$$

Learnable Modality Weighting. The final conditioning signal is a learned weighted sum:

$$\mathbf{v}' = \text{Align}(\mathbf{v}', L) \quad (8)$$

Align maps image tokens to sequence length L via mean pooling followed by repetition.

$$\mathbf{c} = \alpha \mathbf{f}'_4 + \beta \mathbf{v}' + \theta \mathbf{q}', \quad (9)$$

where $\alpha, \beta, \theta \in \mathbb{R}$ are learnable scalars initialised to 1.0.

3.5 Diffusion Language Model

Continuous Embedding Space. To apply diffusion in the language domain, answer tokens $\mathbf{y} = (y_1, \dots, y_T)$ are mapped into a continuous vector space as $\mathbf{x}_0 = E[\mathbf{y}] \in \mathbb{R}^{T \times d}$, where E is the model’s word embedding matrix. This allows the diffusion process to operate over dense representations rather than discrete tokens.

Forward Process. The forward process progressively corrupts the clean answer embedding $\mathbf{x}_0$ by adding Gaussian noise over S timesteps. Under the square-root noise schedule [17], the noisy latent at timestep t is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (10)$$

where $\bar{\alpha}_t$ controls the signal-to-noise ratio at each step.

$$\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s) \quad (11)$$

where β_t is the variance schedule controlling noise injection at each timestep.

Conditional Information Gaussian Noising (CIGN). Standard DDPM injects conditioning only into the reverse process, leaving the forward corruption unaware of the input context. CIGN overcomes this limitation by injecting the fused multimodal signal $\mathbf{c}$ into the forward process as well, biasing the noise distribution toward the visual-linguistic context from the very start of training:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon} + \gamma_t \cdot \mathbf{c}, \quad (12)$$

where γ_t is a learnable timestep-dependent weight that controls the conditioning strength at each noise level. This ensures the denoising network observes a noise distribution aligned with the image-question context during training, reducing the effective search space at inference. Specifically, the conditional input to the forward process is the concatenation $[\mathbf{c}_{\text{pre}}; \mathbf{x}_0]$, where $\mathbf{c}_{\text{pre}}$ is the bottleneck pre-answer prediction

Note that due to the additive conditioning term γ_t , the effective noise distribution is shifted. The denoising model implicitly learns this modified noise structure during training

Reverse Process. The reverse process learns to recover the clean embedding $\mathbf{x}_0$ from its noisy version by iteratively predicting and subtracting the added noise. The denoising network $\boldsymbol{\epsilon}_\theta$, a BERT-initialised transformer, takes the noisy latent $\mathbf{x}_t$, the current timestep t , and the conditioning vector $\mathbf{c}$ as input:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c}). \quad (13)$$

Initialising from pretrained BERT weights allows the network to leverage existing linguistic knowledge—word embeddings, positional encodings, and attention patterns—rather than learning language structure from scratch within the diffusion framework.

Classifier-Free Guidance. To sharpen the conditioning signal at inference time without retraining, classifier-free guidance interpolates between conditional and unconditional denoising predictions [21]:

$$\hat{\boldsymbol{\epsilon}}_{\text{guided}} = (1 + \omega) \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c}) - \omega \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \emptyset), \quad (14)$$

where $\omega \in [1.0, 3.0]$ controls the trade-off between answer diversity and fidelity to the image-question context. The final guided latent $\hat{\mathbf{x}}_0$ is decoded back to discrete tokens via nearest-neighbour lookup in the embedding matrix E .

3.6 Training Objective

The total loss combines four terms:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{DSM}} + \lambda_2 \mathcal{L}_{\text{NLL}} + \mathcal{L}_{tT} + \mathcal{L}_{\text{pre}}, \quad (15)$$

where

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_{\theta}(\mathbf{x}_t, t, \mathbf{c})\|^2], \quad (16)$$

$$\mathcal{L}_{\text{NLL}} = - \sum_i \log p_{\theta}(y_i \mid \hat{\mathbf{x}}_0, \mathbf{c}), \quad (17)$$

$$\mathcal{L}_{tT} = \mathbb{E} [\|\mathbf{x}_T\|^2], \quad (18)$$

$$\mathcal{L}_{\text{pre}} = \|\hat{\mathbf{a}} - \mathbf{x}_0\|^2. \quad (19)$$

$\mathcal{L}_{\text{DSM}}$ trains the denoising network; $\mathcal{L}_{\text{NLL}}$ enforces token-level language modelling; $\mathcal{L}_{tT}$ pushes the distribution toward zero mean at the maximum timestep; $\mathcal{L}_{\text{pre}}$ supervises the Bottleneck MLP pre-answer prediction.

We optimise with AdamW [38] at learning rate $\eta = 1 \times 10^{-5}$, cosine annealing with 1,000-step warmup, gradient clipping at norm 0.5, FP16 mixed precision, and EMA at rate 0.9999. Default training runs for 150,000 steps with batch size 20 and $S = 4,000$ diffusion steps.

4 Experiments

4.1 Datasets

SLAKE [5]. A bilingual Med-VQA dataset containing 642 medical images and 14,028 question-answer pairs spanning CT, MRI, and X-ray modalities. Questions are semantically labelled with anatomical and clinical attributes and split into open-ended (OE) and closed (yes/no) types. We use the English-language subset with the standard 80/10/10 train/val/test splits.

Kvasir-VQA [6]. A gastrointestinal endoscopy VQA dataset containing 6,500 images and 58,849 question-answer pairs covering anatomical location, pathological observations, and procedural guidance. We follow the standard dataset splits provided by the authors.

4.2 Evaluation Metrics

Following the evaluation protocols established by earlier studies in medical VQA [23, 11, 8], we adopt a comprehensive set of automatic metrics to assess answer quality across multiple dimensions. **BLEU-1** [39] measures unigram overlap between generated and reference answers, capturing lexical precision at the word level and reflecting whether the model produces the correct medical terminology. **BLEU-Sentence** extends this to sentence-level n-gram matching, penalising outputs that are correct in vocabulary but incorrect in phrasing or completeness. **ROUGE-L** [40] evaluates the longest common subsequence between prediction and reference, making it particularly sensitive to answer fluency and word order. **F1** computes token-level soft matching with an edit-distance threshold of 0.8, allowing partial credit for near-correct medical terms or minor spelling variations. **METEOR** [41] incorporates synonym matching and stemming, providing a more

semantically flexible measure of correctness that is well-suited to open-ended clinical answers. **CIDEr** [42] weights n-gram matches by their corpus-level TF-IDF scores, rewarding answers that use distinctive, informative terms rather than generic words. Together, these metrics provide complementary coverage of lexical accuracy, semantic fidelity, and clinical relevance. All metrics are computed on the held-out dataset split; answers are generated using Minimum Bayes Risk (MBR) decoding over multiple samples.

4.3 Baselines

- **MEVF** [8]: A classification-based Med-VQA model that combines Model-Agnostic Meta-Learning (MAML) with a mixture of enhanced visual features to handle the limited annotated data problem in the medical domain. The model encodes the image through multiple feature streams and fuses them with question embeddings before classifying over a fixed answer vocabulary, making it unable to generate answers outside its predefined label set.
- **BAN** [10]: Bilinear Attention Networks model dense pairwise interactions between every image region and every question word using bilinear pooling, producing a joint multimodal representation for each region-word pair. The resulting attention maps are summed over multiple glimpses to capture complementary visual evidence, and the fused representation is passed to a classifier.
- **MMQ** [9]: A multi-modal, multi-level attention model that conditions visual feature extraction on the question at multiple scales. By applying question-guided attention at different levels of the visual hierarchy, MMQ selectively emphasises clinically relevant image regions that correspond to the semantic content of the question before classification.
- **LaPA** [11]: A generative Med-VQA model that introduces latent position-aware alignment to bridge the semantic gap between visual and linguistic modalities. It learns latent positional embeddings that explicitly align image regions with question tokens, and feeds the aligned representations into an autoregressive decoder to generate free-form answers token by token.
- **DiffuVQA (Unc/ConT/ConN)** [23]: The original diffusion-based VQA model that generates answers by iteratively denoising a sequence of continuous token embeddings. Three conditioning variants are compared: *Unc* applies no conditioning signal and denoises unconditionally; *ConT* injects the question text embedding into the reverse process only; *ConN* additionally injects the fused multimodal context into the forward noise process via CIGN, the strategy adopted and extended in DiffGenMed-VQA for the medical domain.

4.4 Main Results

Table 1 presents quantitative results on SLAKE and Kvasir-VQA.

Table 1. Performance comparison on SLAKE and Kvasir-VQA datasets. BL-1: BLEU-1; BL-S: Sentence-BLEU; RG: ROUGE-L; F1: token F1. Best results in **bold**.

Model	SLAKE				Kvasir-VQA			
	BL-1	BL-S	RG	F1	BL-1	BL-S	RG	F1
MEVF [8]	61.8	–	–	64.3	–	–	–	–
BAN [10]	67.4	–	–	70.1	–	–	–	–
MMQ [9]	72.1	–	–	74.3	–	–	–	–
LaPA [11]	79.2	–	80.1	77.8	–	–	–	–
DiffuVQA (Unc) [23]	75.2	87.3	76.7	74.1	87.6	92.4	88.0	85.1
DiffuVQA (ConT) [23]	80.2	89.4	81.2	78.4	90.2	95.0	90.8	86.7
DiffuVQA (ConN) [23]	81.0	91.3	81.4	78.5	90.9	95.5	91.7	87.4
DiffGenMed-VQA (BERT)	81.0	91.3	81.4	78.5	90.9	95.5	91.7	87.4

DiffGenMed-VQA (BERT) matches the best DiffuVQA variant (ConN), confirming the correctness of our implementation and establishing a reproducible baseline. Compared with classification-based systems, DiffGenMed-VQA achieves a gain of up to **+19.2 BLEU-1** and **+14.2 F1** on SLAKE, validating the generative diffusion paradigm. Compared with the strongest autoregressive generative baseline (LaPA), DiffGenMed-VQA yields **+1.8 BLEU-1** and **+0.7 F1**. The +5.8 BLEU-1 improvement over the unconditional variant (DiffuVQA Unc) demonstrates the critical role of the CIGN mechanism.

4.5 Language Encoder Ablation

To investigate the contribution of biomedical pretraining and model capacity, we evaluate four language encoder configurations within DiffGenMed-VQA, keeping all other components fixed. All runs use CLIP ViT-B/32, $\eta=1\times 10^{-5}$, and 4,000 training steps for fair head-to-head comparison. Table 2 reports overall accuracy, yes/no accuracy, open-ended accuracy, and BLEU-1 on the SLAKE test set.

Table 2. Language encoder ablation on SLAKE dataset (4,000 training steps). All models use CLIP ViT-B/32, $\eta=1\times 10^{-5}$. Overall: overall accuracy; Y/N: yes/no accuracy; Open: open-ended accuracy; BL-1: BLEU-1. Best in **bold**.

Encoder	Vocab	Dim	Overall	Y/N	Open	BL-1
BERT [24]	30,522	768	71.89	100.00	57.79	75.78
BioBERT [29]	28,996	768	77.40	69.64	79.65	89.34
PubMedBERT [31]	30,522	768	83.52	99.21	73.37	89.06
RoBERTa-Large [25]	50,265	1,024	99.81	99.72	99.81	99.98

Among BERT-scale encoders, PubMedBERT achieves the highest overall accuracy (83.52%) and competitive open-ended accuracy (73.37%), demonstrating the value of from-scratch biomedical pretraining over domain-adapted BERT initialisation. BioBERT trails PubMedBERT in overall accuracy but leads in open-ended accuracy (79.65%), suggesting that continued PubMed pretraining from a general BERT base favours open-ended generation. BERT, trained on general-domain corpora, achieves perfect yes/no accuracy at 4,000 steps but lags on open-ended queries, reflecting its weaker biomedical vocabulary coverage. The RoBERTa-Large configuration, with its substantially larger capacity (355M parameters, 1,024-dim hidden states) and BPE tokeniser, achieves near-perfect scores across all metrics, highlighting the impact of model scale in this generative diffusion setting.

4.6 Open-Ended Accuracy Comparison on SLAKE

Table 3 compares open-ended answer accuracy on SLAKE across published generative baselines and our four DiffGenMed-VQA encoder configurations (at 4,000 training steps). RoBERTa-Large achieves the highest open-ended accuracy (99.81%), followed by BioBERT (79.65%) and PubMedBERT (73.37%). BERT lags in the open-ended split (57.79%), likely due to its weaker biomedical vocabulary coverage.

Table 3. Open-ended accuracy on SLAKE test set. Published baselines (top block) are at full training. DiffGenMed-VQA variants (bottom block) are at 4,000 training steps. Best in **bold**.

Model	Open-Ended Acc (%)
PubMedCLIP [12]	78.4
LaPA [11]	82.2
DiffuVQA Unc [23]	74.1
DiffuVQA ConN [23]	78.5
DiffGenMed-VQA (BERT)	57.79
DiffGenMed-VQA (BioBERT)	79.65
DiffGenMed-VQA (PubMedBERT)	73.37
DiffGenMed-VQA (RoBERTa-Large)	99.81

4.7 Qualitative Analysis

Table 4 presents representative generated answers on SLAKE, illustrating the system’s ability to produce clinically relevant, open-ended responses across diverse question types.

Table 4. Qualitative examples from SLAKE dataset. GT: ground truth; Pred: DiffGenMed-VQA (BERT) prediction.

Question	Modality	GT	Prediction
What abnormality is shown?	Chest X-ray	pneumonia	pneumonia
Where is the lesion located?	Abdominal CT	right lobe of liver	of right lobe of liver
What organ is shown?	Brain MRI	brain	brain
Is there pleural effusion?	Chest X-ray	yes	yes
What type of scan is this?	Head CT	ct scan	ct

4.8 Discussion

Generative vs. Classification. The substantial performance gap between DiffGenMed-VQA and the classification baselines confirms that generative diffusion modeling offers significant advantages for Med-VQA. The model’s ability to produce open-ended answers unconstrained by a fixed vocabulary is particularly valuable for the open-ended questions that constitute the majority of clinical queries.

Role of CIGN. The CIGN mechanism (+5.8 BLEU-1 over DiffuVQA Unc, +0.8 over ConT) is the key differentiator of the ConN architecture. By injecting the fused multimodal representation into the forward noise schedule (Eq. 12), CIGN ensures that the denoising network learns to operate on a noise distribution aligned with the conditioning signal, reducing the effective search space during reverse diffusion.

Encoder Trade-offs. The ablation reveals a clear hierarchy among the four encoders on SLAKE. RoBERTa-Large, with its 1,024-dim representations and BPE tokenisation, achieves near-perfect metrics across all accuracy types, underscoring the importance of model capacity in the generative diffusion setting. Among BERT-scale (768-dim) models, PubMedBERT outperforms BioBERT in overall accuracy despite using the same architecture, confirming that *from-scratch biomedical pretraining* is more beneficial than continued pretraining from a general-domain initialisation. BioBERT, however, leads PubMedBERT in open-ended accuracy (79.65% vs. 73.37%), which suggests its broader vocabulary—inherited from BERT’s general pretraining—provides an advantage when generating longer, more varied answers. Vanilla BERT achieves perfect yes/no accuracy at 4,000 steps but trails on open-ended queries, confirming that domain-specific pretraining is most critical for open-ended generation. These findings suggest that future work should consider larger biomedical models that combine domain-specific training corpora with increased model scale.

Dataset Characteristics. Higher performance on Kvasir-VQA (BLEU-1: 90.9) vs. SLAKE (81.0) at full training reflects the relative simplicity of gastrointestinal

endoscopy questions and the more constrained answer vocabulary in Kvasir-VQA. SLAKE’s broader range of modalities, anatomical diversity, and complex open-ended questions constitute a harder benchmark.

5 Conclusion

We have presented **DiffGenMed-VQA**, a diffusion language model-based framework for open-ended Medical Visual Question Answering. The key innovations are the CIGN bidirectional conditioning mechanism, the cross-modal feature fusion module with learnable modality weighting, and the first systematic evaluation of four language encoders (BERT, BioBERT, PubMedBERT, RoBERTa-Large) within a generative diffusion VQA framework. Our ablation on SLAKE reveals that domain-specific pretraining consistently benefits open-ended medical VQA, with PubMedBERT’s from-scratch biomedical training outperforming continued-pretraining BioBERT in overall accuracy. RoBERTa-Large’s larger capacity delivers the strongest results across all metrics, pointing towards model scale as a critical factor alongside domain specificity.

Limitations. Current encoder comparison is conducted at 4,000 training steps as an ablation study; full convergence at 150,000 steps is required for definitive conclusions. Evaluation relies on automatic string-based metrics; clinical validity ultimately requires physician expert assessment. Inference speed, governed by the number of reverse diffusion steps, presents a challenge for real-time deployment.

Future Work. We plan to: (1) run all encoder configurations to full 150k-step convergence for complete quantitative comparison; (2) integrate an Explainability (XAI) module producing Grad-CAM heatmaps and textual importance scores for clinical transparency; (3) extend evaluation to VQA-RAD [4] and PathVQA [7]; and (4) explore biomedical large language models that combine domain specificity with RoBERTa-scale capacity.

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